

Peter Juhasz — Researcher

✉ peter.juhasz@math.au.dk • 🌐 www.peter-juhasz.com • 🎓 Google Scholar
🆔 ORCID • 📄 ResearchGate • in LinkedIn • 🐙 GitHub

Summary

- PhD Fellow in Mathematics
- Completed research projects in spatial statistics, network science, automated driving and complex systems
- Published multiple research papers and patented a Gaussian regression based trajectory prediction model

Education

- **PhD in Mathematics**
Aarhus University, Denmark 2022–
Thesis: Topological Data Analysis Based Models of Evolving Higher-Order Networks
- **Master of Business Administration with Highest Honors**
Budapest University of Technology and Economics, Hungary, Specialized in Finance 2019–2021
Thesis: Prediction of Popularity of Memes using Machine Learning Methods
- **Master of Science in Physics**
Budapest University of Technology and Economics, Hungary, Specialized in Applied Physics 2013–2016
Thesis: Examination of Spatially Embedded Complex Networks
Erasmus+ Scholarship: Delft University of Technology (Applied Physics)
- **Bachelor of Science in Physics**
Budapest University of Technology and Economics, Hungary, Specialized in Applied Physics 2010–2013
Thesis: Finite Element Method Plasma Simulation of Nitrogen Contaminated Ceramic Metal Halide Lamps

Professional Experience

- **Research Assistant in Spatial Statistics and Topological Data Analysis**
Department of Mathematics, Aarhus University, Denmark 2022–
 - Developed topological data analysis based models for higher-order networks
 - Proved theorems in the field of marked spatial point processes
- **Deep Learning Researcher in Automated Driving**
Department of Automated Driving, Bosch Group Hungary 2021–2022
 - Trajectory prediction with Gaussian regression
 - Real time object tracking with temporal convolutional neural networks
 - Object detection and classification with ultrasonic sensors
- **Software Architect in Automated Driving**
Department of Automated Driving, Bosch Group Hungary 2018–2021
 - Coordinated 2 teams in the integration of driver monitoring camera for hands free driving
 - Certified Professional in Software Architecture (International Software Architecture Qualification Board)
- **Software Engineer for Mobile Networks**
Business Unit IT & Cloud Products, Ericsson Telecommunications Hungary 2016–2018
 - Analyzed response times of distributed database servers for optimal load balancing
 - Coordinated a team of eight as a deputy of the product owner
 - Implemented performance critical algorithms for Smart Services Routers

Publications, Patents

- Barnes, K., Juhasz, P., Nagy, M. & Molontay, R. (2024).
From NYT Headlines to Reddit Memes: Topicality Boosts Meme Popularity. 2024
Social Network Analysis and Mining, 12(1), 120–121.
<https://doi.org/10.21203/rs.3.rs-3959722/v1>
- Hirsch, C., Jahnel, B., Jhawar, S. K. & Juhasz, P. (2023).
Poisson approximation of fixed-degree nodes in weighted random connection models. 2023
arXiv preprint, Article 2311.12643.
<https://doi.org/10.48550/arXiv.2311.12643>
- Hirsch, C. & Juhasz, P. (2023).
On the topology of higher-order age-dependent random connection models. 2023
arXiv preprint, Article 2309.11407.
<https://doi.org/10.48550/arXiv.2309.11407>
- Juhasz, P. (2021).
Information propagation in stochastic networks. 2021
Physica A: Statistical Mechanics and its Applications, 577 Article 126070.
<https://doi.org/10.1016/j.physa.2021.126070>
- Juhasz, P. (2020).
Method and device for predicting the trajectory of a traffic participant, and sensor system. 2020
(Patent No. CNIPA: CN113988353A, EPO: EP3916697A1, JPO: JP2021190119A, USPTO: US2021366274A1).
<https://patents.google.com/patent/US20210366274A1>
- Juhasz, P., Stéger, J., Kondor, D. & Vattay, G. (2018).
A Bayesian approach to identify Bitcoin users. 2018
Public Library of Science One, 13(12), Article e0207000.
<https://doi.org/10.1371/journal.pone.0207000>
- Juhasz, P., Beleznai, Sz. & Maros, I. (2014).
Finite element method plasma simulation of nitrogen contaminated metal halide lamps. 2014
Proceedings of the 2014 Comsol Conference, Article .
https://www.comsol.com/paper/download/199945/juhaasz_paper.pdf

Conferences, Presentations

- **Poster: Topological Data Analysis of Higher-Order Networks** 2023
Danish–Swedish Summer School on TDA and Spatial Statistics
- **Presentation: Trajectory Prediction Using a Gaussian Regression Model** 2020
Bosch Group: Bosch Innovation Day
- **Presentation: Probabilistic Localization of Bitcoin Users** 2016
Hungarian Academy of Sciences: Statistical Physics Conference
- **Poster: Dissecting Distributed Database Systems For Telecom Applications** 2016
Ericsson Telecommunications Hungary: Ericsson University Day
- **Poster: Finite Element Method Plasma Simulation of Ceramic Metal Halide Lamps** 2014
Comsol Multiphysics Conference

Teaching Experience

- **Data Project**
Aarhus University, Course Code: F24.550201U004.A 2024
10 ECTS Computer Project Class, Spring Semester
- **Linear Transformations**
Aarhus University, Course Code: E23.550171U010.A 2023
10 ECTS Exercise Class, Autumn Semester
- **Ordinary Differential Equations**
Aarhus University, Course Code: F23.550141U011.A 2023
5 ECTS Exercise Class, Spring Semester
- **Advanced Calculus for Engineers**
Aarhus University, Course Code: F23.280191U021.A 2023
10 ECTS Exercise Class, Spring Semester

Awards, Grants

- **Academic Mobility Grant for International Exchange** 2024–2025
Evolving Stochastic Network Models
20 000 DKK, Aarhus University
- **PhD Fellowship Grant DDSA-PhD-2022-008** 2022–2025
Topological Data Analysis Based Models of Evolving Higher-Order Networks
1 890 000 DKK, Danish Data Science Academy
- **Excellence Award for Outstanding Thesis and Diploma Work** 2021
Digitalization, Artificial Intelligence, and the Age of Data – What Makes a Meme Viral?
350 000 HUF, Hungarian National Bank

Programming Skills

- **Python** ● ● ● ● ●
- **Git** ● ● ● ● ●
- **C++** ● ● ● ● ●

Languages

- **Hungarian** ● ● ● ● ●
- **English** ● ● ● ● ●
- **German** ● ● ● ● ●